Doctorate Research Project

Extreme heat and its impact in hospitalizations and mortality in Rio de Janeiro: risk levels and sociospatial disparities

João Henrique de Araujo Morais
Advisor: **Oswaldo G. Cruz**
Co-advisor abroad: **Rachel Lowe**

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FIOCRUZ

Epidemiology in Public Health Doctorate Program
Sergio Arouca National School of Public Health (ENSP/Fiocruz)
March 2026

CIE
CENTRO DE INTELIGÊNCIA
EPIDEMIOLÓGICA

Contents

- Research questions
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- What is to be done
  - Spatial Bayesian DLNMs
  - Spatial level of aggregation
  - Accounting for differences in exposure

Research questions

- Which causes of death, hospitalization and emergency visits are most associated with extreme heat events in Rio de Janeiro city, by age group?

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Done | Considering | To-do

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Done Considering To-do

What has been done

- Preprint article:

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João Henrique de Araujo Morais, Débora Medeiros de Oliveira e Cruz, Valeria Saraceni, Caroline Dias Ferreira, Gislani Mateus Oliveira Aguilar, Oswaldo Gonçalves Cruz

doi: https://doi.org/10.1101/2025.01.17.25320740

This article is a preprint and has not been peer-reviewed [what does this mean?]. It reports new medical research that has yet to be evaluated and so should *not* be used to guide clinical practice.

Abstract | **Full Text** | Info/History | Metrics | Preview PDF

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Environmental Epidemiology

ISEE ENVIRONMENTAL EPIDEMIOLOGY

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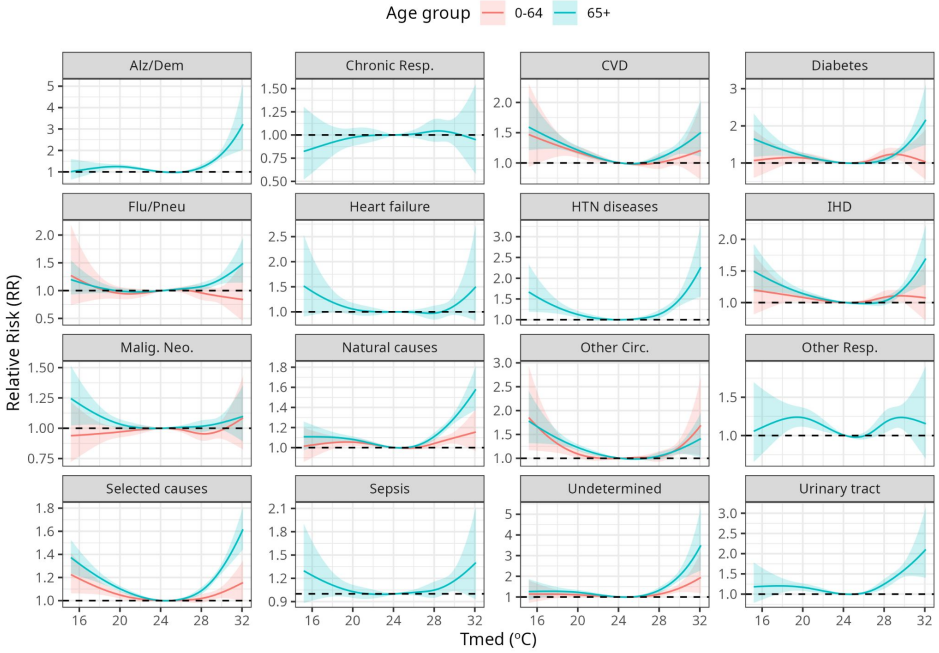
  - Splitted into two papers

What has been done



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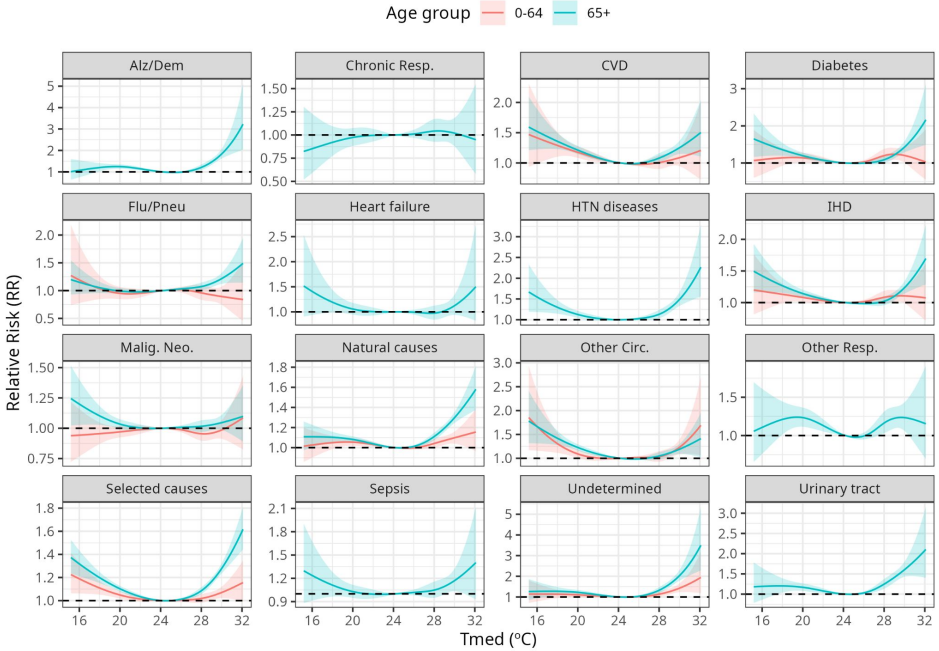
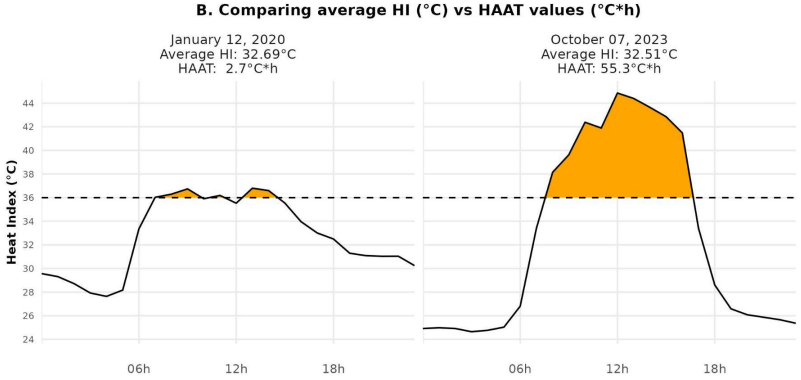


- Proposal of a novel metric **Heat Area Above a Threshold (HAAT)**



What is to be done

- Inclusion of **spatial component** in the DLNM models for deaths

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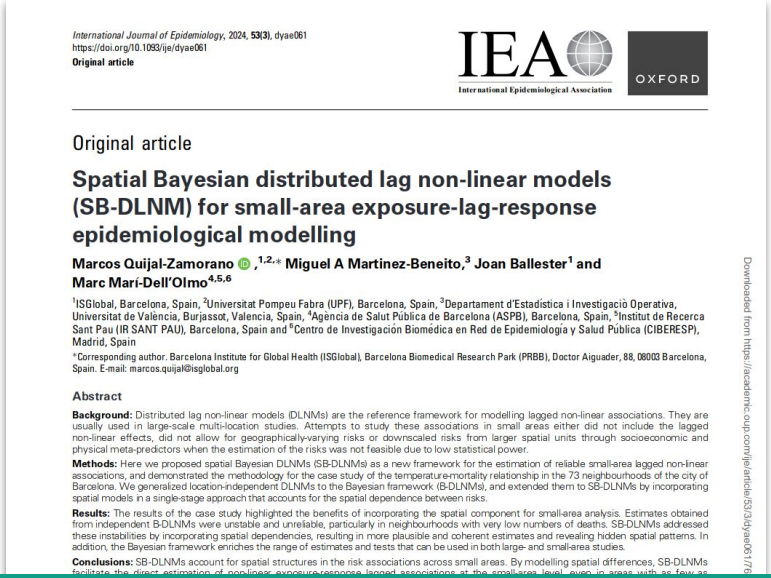

International Journal of Epidemiology, 2024, **53(3)**, dyae061
https://doi.org/10.1093/ije/dyae061
Original article

IEA International Epidemiological Association

OXFORD

Original article

Spatial Bayesian distributed lag non-linear models (SB-DLNM) for small-area exposure-lag-response epidemiological modelling

Marcos Quijal-Zamorano,[1,2,*] **Miguel A Martinez-Beneito**,[3] **Joan Ballester**[1] **and Marc Marí-Dell'Olmo**[4,5,6]

[1]ISGlobal, Barcelona, Spain, [2]Universitat Pompeu Fabra (UPF), Barcelona, Spain, [3]Departament d'Estadística i Investigació Operativa, Universitat de València, Burjassot, Valencia, Spain, [4]Agència de Salut Pública de Barcelona (ASPB), Barcelona, Spain, [5]Institut de Recerca Sant Pau (IR SANT PAU), Barcelona, Spain and [6]Centro de Investigación Biomédica en Red de Epidemiología y Salud Pública (CIBERESP), Madrid, Spain

*Corresponding author. Barcelona Institute for Global Health (ISGlobal), Barcelona Biomedical Research Park (PRBB), Doctor Aiguader, 88, 08003 Barcelona, Spain. E-mail: marcos.quijal@isglobal.org

Abstract

Background: Distributed lag non-linear models (DLNMs) are the reference framework for modelling lagged non-linear associations. They are usually used in large-scale multi-location studies. Attempts to study these associations in small areas either did not include the lagged non-linear effects, did not allow for geographically-varying risks or downscaled risks from larger spatial units through socioeconomic and physical meta-predictors when the estimation of the risks was not feasible due to low statistical power.

Methods: Here we proposed spatial Bayesian DLNMs (SB-DLNMs) as a new framework for the estimation of reliable small-area lagged non-linear associations, and demonstrated the methodology for the case study of the temperature-mortality relationship in the 73 neighbourhoods of the city of Barcelona. We generalized location-independent DLNMs to the Bayesian framework (B-DLNMs), and extended them to SB-DLNMs by incorporating spatial models in a single-stage approach that accounts for the spatial dependence between risks.

Results: The results of the case study highlighted the benefits of incorporating the spatial component for small-area analysis. Estimates obtained from independent B-DLNMs were unstable and unreliable, particularly in neighbourhoods with very low numbers of deaths. SB-DLNMs addressed these instabilities by incorporating spatial dependencies, resulting in more plausible and coherent estimates and revealing hidden spatial patterns. In addition, the Bayesian framework enriches the range of estimates and tests that can be used in both large- and small-area studies.

Conclusions: SB-DLNMs account for spatial structures in the risk associations across small areas. By modelling spatial differences, SB-DLNMs facilitate the direct estimation of exposure-response lagged associations at the small-area level, even in areas with as few as

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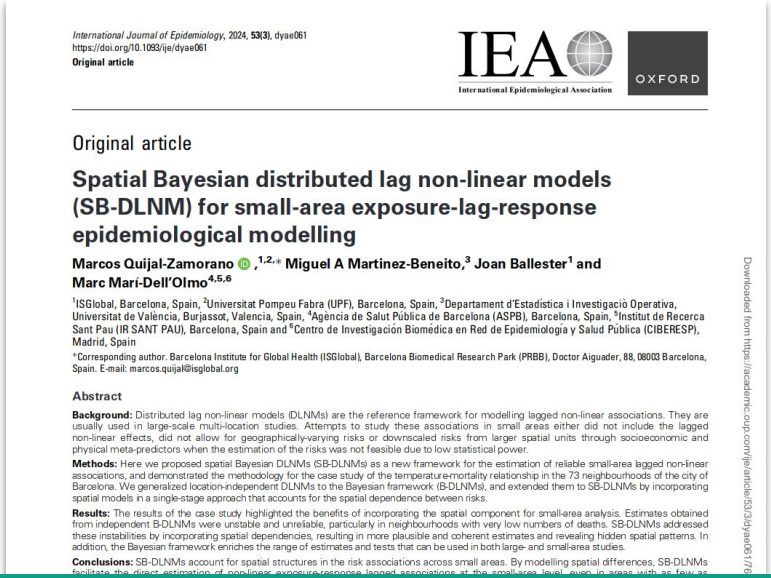

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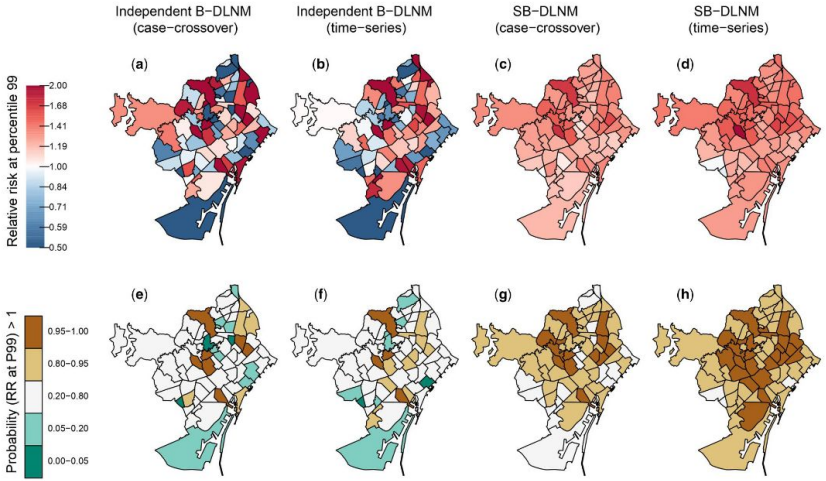

Extracted from: QUIJAL-ZAMORANO, Marcos et al. Spatial Bayesian distributed lag non-linear models (SB-DLNM) for small-area exposure-lag-response epidemiological modelling. International Journal of Epidemiology, v. 53, n. 3, p. dyae061, 11 abr. 2024.

What is to be done: three issues...

1. Inclusion of:

   i) non-linearity of exposure-response relationship;

   ii) lagged effects;

   iii) spatially structured dependence among areas;

   iv) role of spatial covariates

is too complex of a model (until iii it has been done)**.**

What is to be done:

Proposal: **two-stage modeling**

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- From posterior distributions of the exposure-response surface obtained for each area:
  - Use a specific metric (eg.: exceedance probabilities) as the outcome in **a second spatial model,** with covariates
  - Estimate the role of desired covariates on explaining risk variability obtained in the first stage

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Proposal: **two-stage modeling**

- First stage: Standard Spatial Bayesian DLNM (SB-DLNM) as proposed by Quijal-Zamorano;
- From posterior distributions of the exposure-response surface obtained for each area:
  - Use a specific metric (eg.: exceedance probabilities) as the outcome in **a second spatial model,** with covariates
  - Estimate the role of desired covariates on explaining risk variability obtained in the first stage
- Need to incorporate uncertainty in the outcome somehow;

What is to be done:

In Quijal-Zamorano's study, execution times were much lower for Case-crossover designs (no trend and seasonality terms):

Instead of adding seasonality and long-term trend components in SB-DLNM, <u>an expected value of deaths could be calculated</u> for each area in each day beforehand, in a simpler model

- Would also need to propagate uncertainty

MODEL	DESIGN	EXECUTION TIME WinBUGS	EXECUTION TIME R-INLA EXECUTION TIME	RATIO WinBUGS / R-INLA
Independent B-DLNM	Case-crossover	>2 days	35 seconds	~ 5.000
Independent B-DLNM	Time-series	>3 days	11 minutes	~ 400
SB-DLNM	Case-crossover	>14 days	25 minutes	~ 800
SB-DLNM	Time-series	>14 days	20.1 hours	~ 17

Extracted from: QUIJAL-ZAMORANO, Marcos. Temperature impacts on mortality at different spatial and temporal scales. Ph.D. Thesis]: Universitat Pompeu Fabra, 14 nov. 2024.

What is to be done: three issues...

2. Spatial level of aggregation

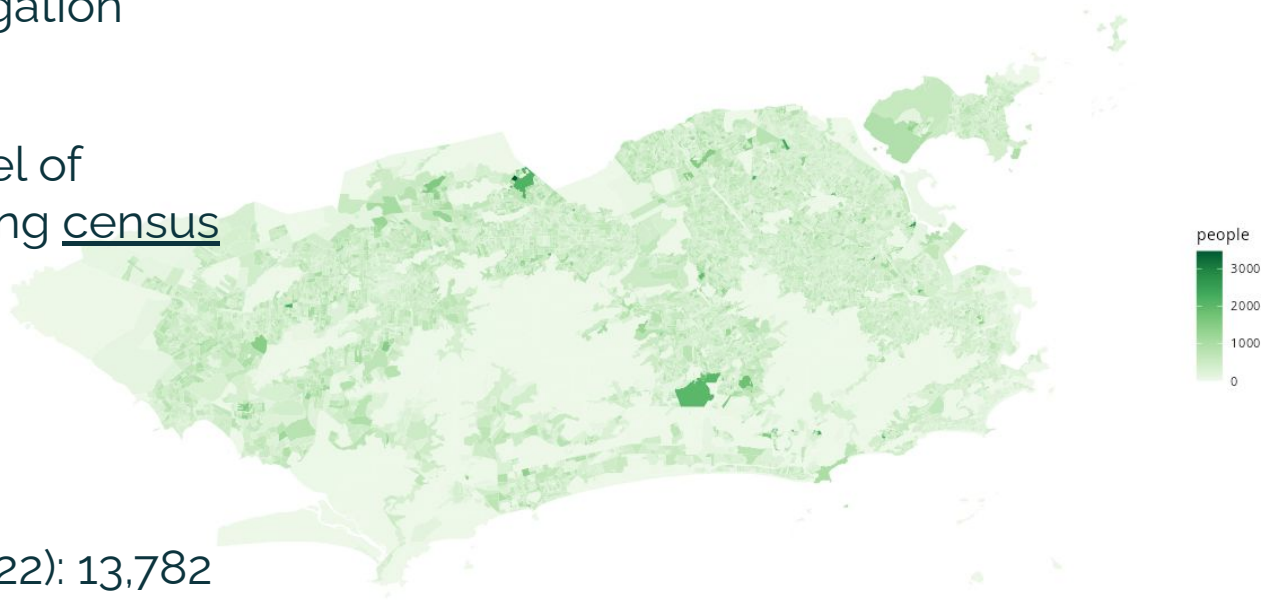
Admistritative regions (RAs)
n=33, too broad

Neighborhoods
n=166, too heterogeneous
Pop. ranging from ~1k to ~350k
Some few counts, some too broad

What is to be done: three issues...

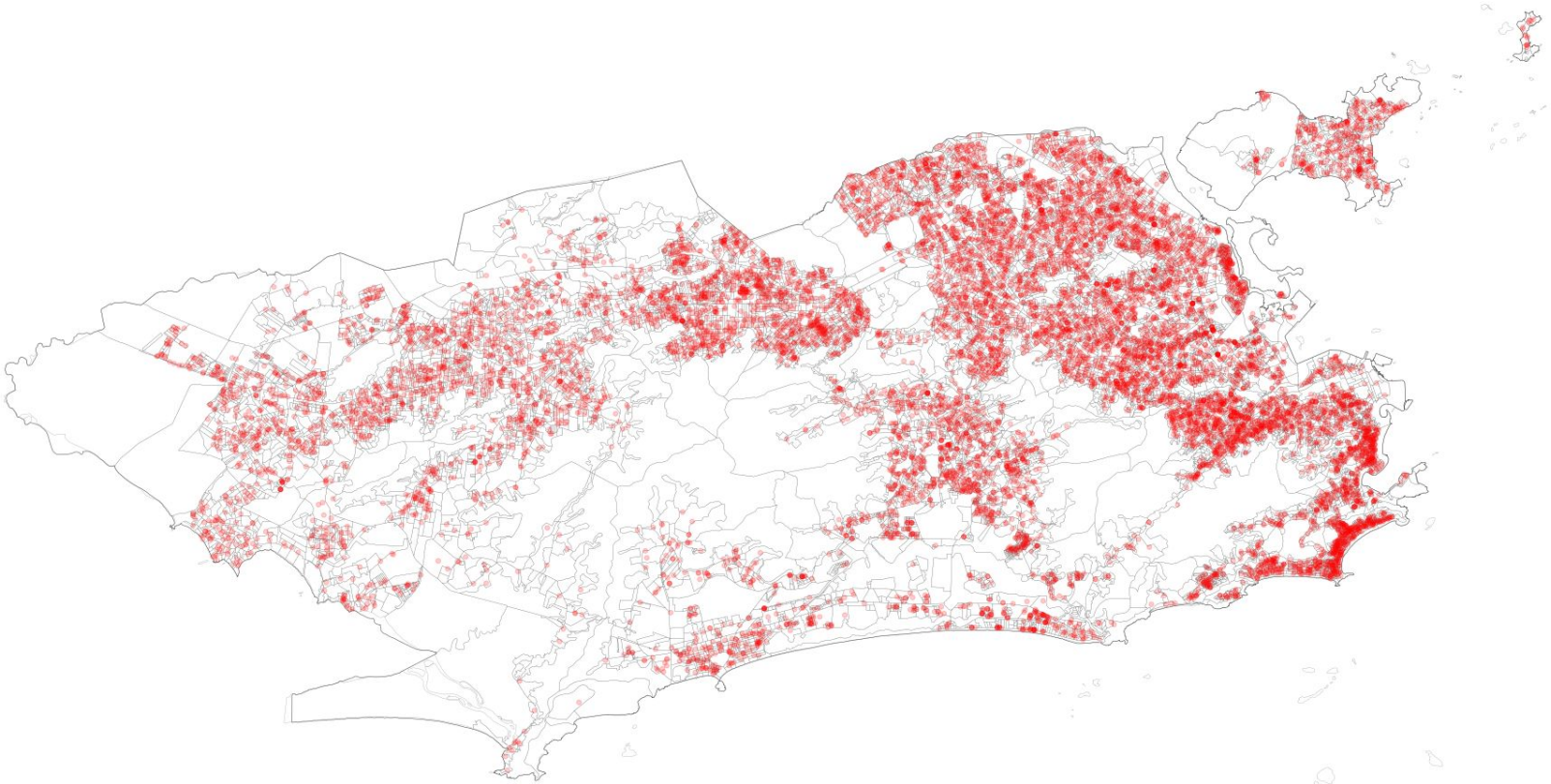
2. Spatial level of aggregation

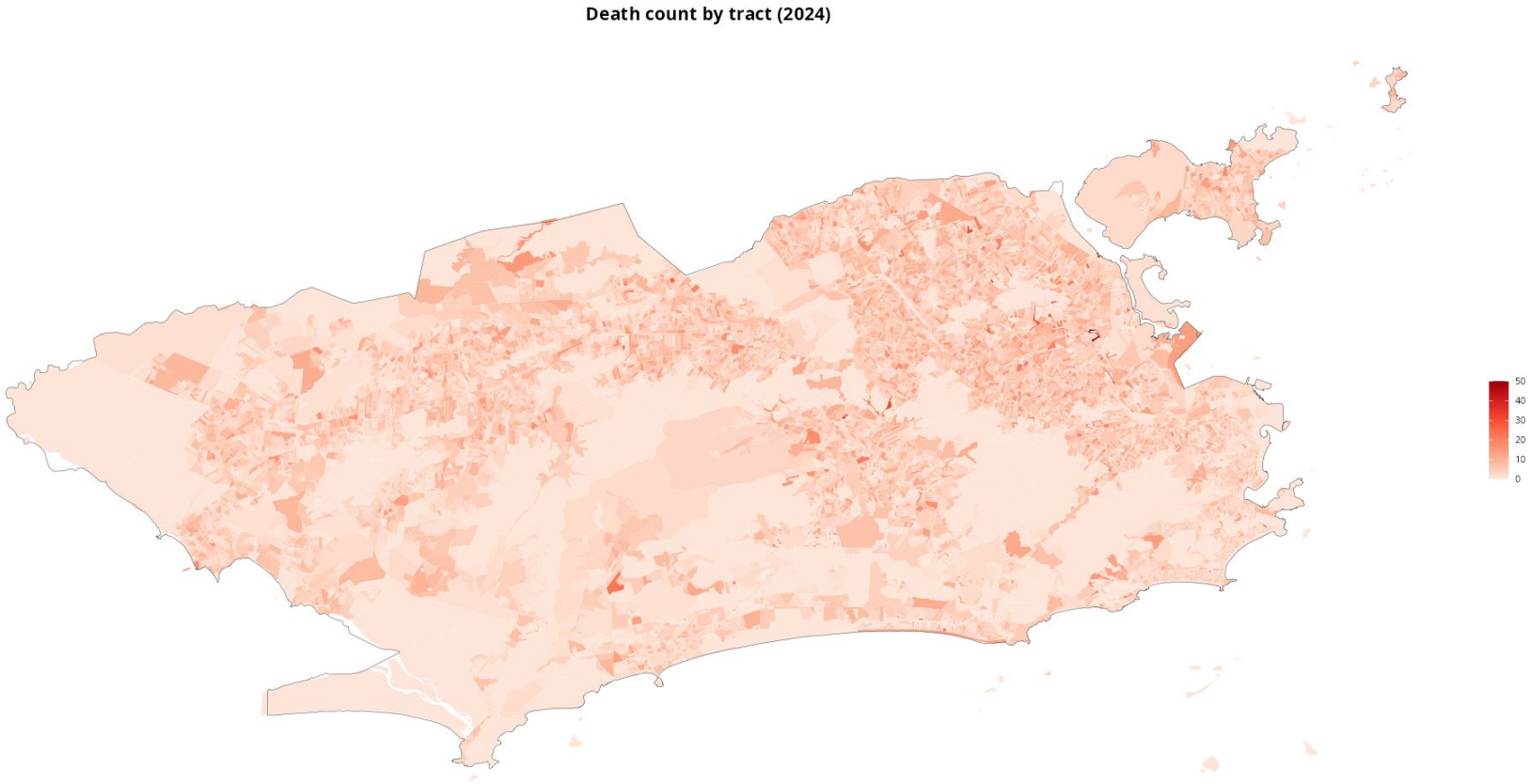
Idea: intermediate level of aggregation by grouping census tracts



N° of census tracts (2022): 13,782

Geocoded deaths




Death count by tract (2024)
50
40
30
20
10
0

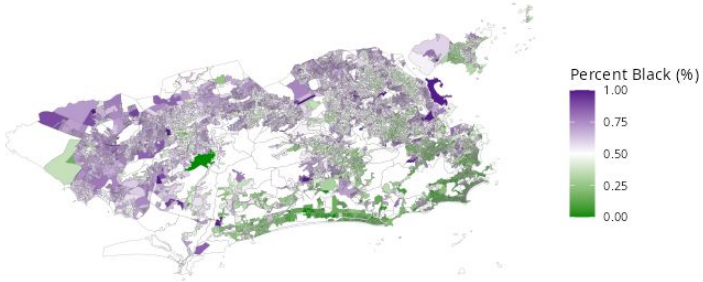
What is to be done: three issues...

General algorithm idea for spatial clustering

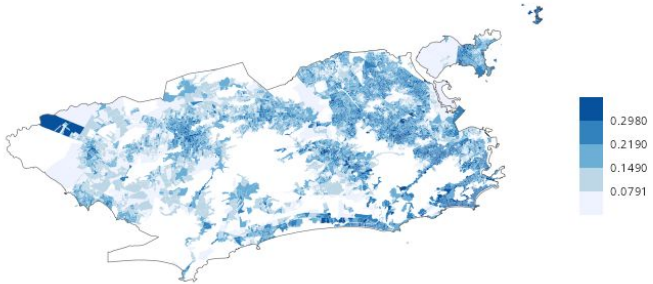
```
while (exists region with number of deaths < DEFINED_THRESHOLD and area_km2
< AREA_LIMIT) do:

     pick area a with lowest deaths;
     N_a <- neighbors(a);
     find neighbor n in N_a that is most similar to a according to
covariates;
     join(a,n);
     update adjancy matrix, number of deaths and covariates;

end
```

Covariates from census (2022)

Racial distribution (2022)

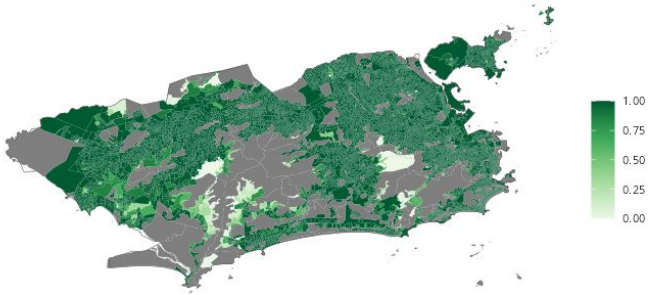


Percent elderly

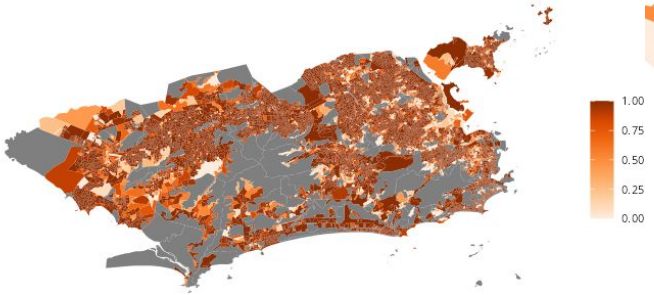


Percent illiterate

Percent water treatment

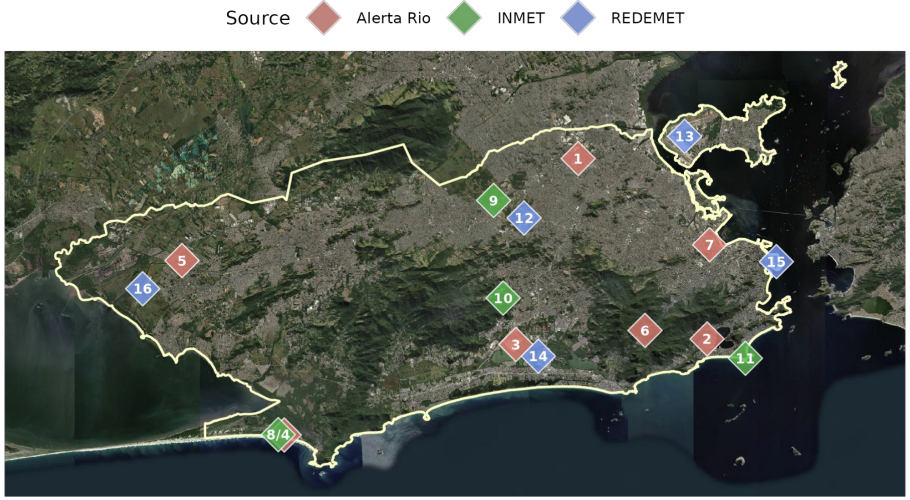


Percent trash collection



What is to be done: three issues...

3. Spatial difference in exposure



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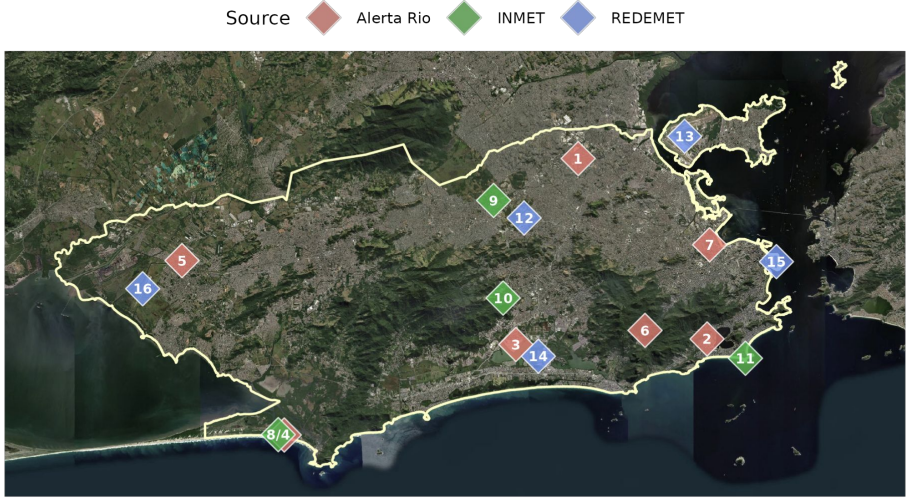
3. Spatial difference in exposure

- Not simple, mainly because:
  - Simpler approaches would be too unreliable given station heterogeneity



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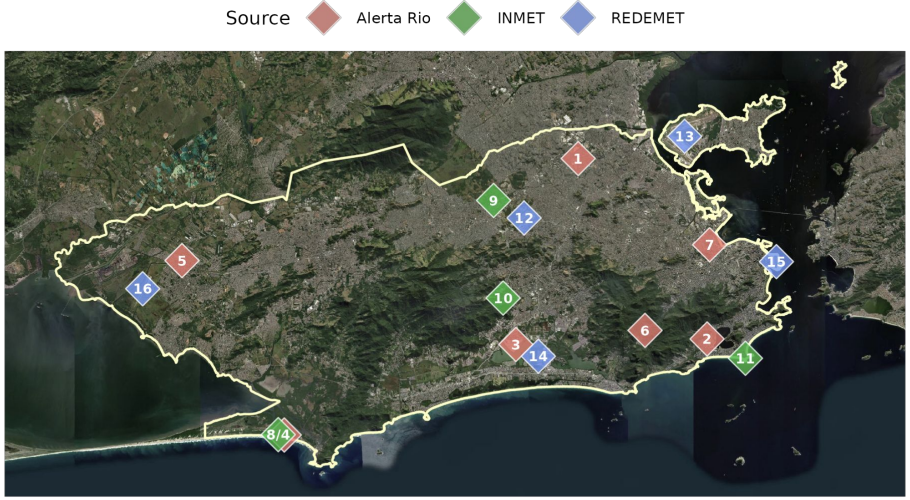
3. Spatial difference in exposure

- Not simple, mainly because:
  - Simpler approaches would be too unreliable given station heterogeneity
- In principle, not the focus of the study
- Objective is to estimate spatial differences in risk, not in exposure



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Interpretation would be: "When Rio's median temperature reaches X ˚C, risk of mortality by natural causes increases by R_1% in area 1, R_2% in area 2, etc...

Thank you!

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